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A Novel Channel Selection Method for BCI Classification Using Dynamic Channel Relevance

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ABSTRACT Brain-Computer Interface (BCI) provides a direct communicating pathway between the human brain and the external environment. In the BCI systems, electroencephalography (EEG) signals are used to represent different cognitive patterns corresponding to various limb movements or motor imagery (MI) activities. However, EEG signals are multichannel in nature that require explicit information processing to alleviate the computational complexity of the BCI system. This paper represents a novel channel selection method to find effective EEG channels using Dynamic Channel Relevance (DCR) score. The proposed approach explores the minimum redundancy maximum relevance paradigm for selecting pertinent channels from raw ones. Firstly, the EEG signals are preprocessed using the Savitzky-Golay filter, and a sliding window approach is used to decompose them into chunks of fixed length. Further, three priorly known channels are used as candidate solutions for detecting new correlated channels. The selected channels are used to extract spatial-temporal features using the Multivariate Empirical Mode Decomposition (MEMD) method. The extracted features are used to discriminate four MI tasks (left hand, right hand, tongue, and foot) using the Support Vector Machine (SVM). The experiment is validated on three public EEG datasets (BCI Competition IV- 2008 – IIA, BCI Competition IV- dataset 1, BCI competition III - dataset IVa). The results show that the proposed method achieved a superior classification accuracy (85.4% on dataset 1, 80.33% on dataset 2, and 85.20% on dataset 3) with a lesser number of channels compared to state-of-the-art methods. In addition, our method significantly reduced the computation time compared to other published results without compromising the classification accuracy. Topographical mapping between the selected channels and cognitive regions showed that the central, frontal, and parietal lobe contributes to the execution of various MI tasks during physical activities.

INDEX TERMS Brain-Computer interface, electroencephalography, dynamic channel relevance, multivariate empirical mode decomposition, support vector machine.

I. INTRODUCTION

Brain-Computer Interface (BCI) is an emerging technology that provides an alternative communication pathway between the human brain and the external world. It intakes neural signals from the human brain, analyses, and translates them into computer-based control commands that regulate an external prosthetic device to perform a desired activity [15]. Common BCI systems are based on various neuroimaging techniques such as electroencephalogram (EEG) and magnetoencephalogram (MEG) [1], [12], [19]. The primary objective of newly developed BCI systems is to improve communication ability between physically disabled persons and different prosthetic devices such as incontinence control machines, hearing aids, and robotic limbs [29].

In EEG-based BCI communication, Motor Imagery (MI) is an important paradigm [24], [40] that deals with imagination-based muscular activities. A subject is required to imagine in his brain about a limb movement without performing any physical task. However, the implementation of MI-based BCI systems is notoriously difficult because their training procedure is time-consuming and shows limited BCI channel capacity. For research purposes, EEG signals may be recorded from the scalp during the MI tasks so that

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practitioners can understand the dynamics of the complex brain activities during MI activities' execution.

In general, EEG signals are non-stationary, chaotic, and multichannel in nature [18], [42]. In EEG signal processing, MI-specific signals are collected from definite scalp sites referred to as channels or electrodes. Although the dense arrangement of electrodes on the human scalp explores more information about the underlying mechanism of neuronal activity but induces an abundant amount of redundant and irrelevant information in the form of noise and high-dimensional data. In addition, it increases the hardware complexity of the BCI system, which requires more effort during the BCI preparation setup. Therefore, it is essential to reduce these efforts using a minimal but most informative set of channels.

The human brain consists of different regions associated with various tasks; for instance, the motor cortex with the planning and various Motor Imagery (MI) specific voluntary movements, the visual cortex for visual processing, and the Pre-Frontal Cortex (PFC) along with the hippocampus perform decision-making tasks [62]. Thus, the identification of MI task-related locations is a preliminary step performed during BCI setup installation. In addition, this step decreases the computational complexity and minimizes the overfitting problem during the BCI training sessions. In this study, multichannel EEG signal processing for MI task discrimination is addressed as a multiclass classification problem, as shown in Fig. 1.

Several attempts have been made to solve the optimal channel selection problem by correlating limb movements with respective cognitive patterns. In these experiments, prior information about neural correlation paved the way to select MI task-related channels from the motor cortex and its neighboring regions. This approach often yields a set of highly significant channels treated as candidate solutions that determine other relevant electrodes. To date, several information-theoretic concepts such as normalized mutual information (NMI) [6], entropy [16], correlation coefficient [33], and chi-squared statistics [8] have been used to evaluate the relevance of correlated channels.

In this paper, a novel channel selection method named Dynamic Channel Relevance (DCR) is proposed to discriminate multiple MI tasks such as left hand, right hand, tongue, and feet. The proposed approach evaluates the relevance of a channel in terms of two factors: (1) minimum redundancy and (2) maximum importance. It dynamically selects channels according to their high importance and low redundancy score. This method is motivated by the Dynamic Feature Importance (DFI) [65] approach that combines Gini Importance (GI) [47] and Mutual Information Coefficient (MIC) [44] to measure the importance and redundancy of a selected channel. Here, we first performed EEG preprocessing to reduce the noise and outliers from the raw signals. The preprocessed EEG signals are decomposed into equal size chunks using the sliding window method [30], and later the Multivariate Empirical Mode Decomposition (MEMD) [24] is adopted for feature extraction.

Further, new channels are selected using the DCR index, and respective features are used in the classification step. A Support Vector Machine (SVM) classifier [32], [63] is used to discriminate multiple MI tasks. Finally, topographical maps are plotted for all the selected channels, and related cognitive regions are correlated with the performed MI tasks. The objectives of this paper can be summarized as follows:

1. To study the variance among different cognitive patterns corresponding to four MI tasks.
2. We propose a channel relevance index to rank selected channels based on their importance and redundancy level associated with the candidate channels.
3. To develop a multiclass classification model to discriminate four MI tasks using the subject and the channel-specific MEMD features.
4. To present the results in terms of Channel Reduction Rate (CRR) and classification accuracy (CA), Specificity, and Sensitivity when EEG chunks are discriminated against the extracted features.
5. To find the correlation between MI tasks and the different cognitive regions recognized by the topographical map of selected channels.

The remaining paper is structured as follows. In section II, we present the chronological overview of the background research work that prompted our study. Section III explains the proposed channel selection strategy, while the results are discussed in section IV. The discussion of the proposed research work is recapitulated in section V and finally, the conclusion of the work is listed in section VI.

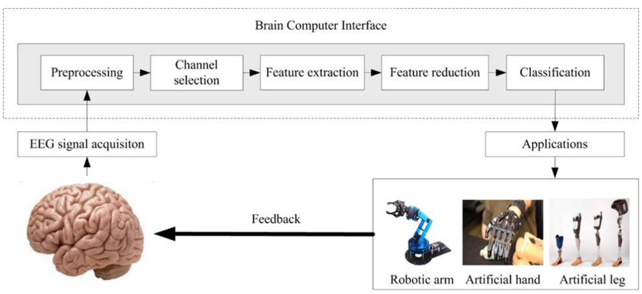

FIGURE 1. A generalized BCI classification framework.

II. RELATED WORK

In optimization theory, EEG channel selection can be considered a feature reduction problem. Therefore, several feature extraction methods have been explicitly employed to solve the optimal channel selection problem. A detailed study of existing channel selection methods is given in [8]. These approaches can be categorized into three groups: (1) Filter methods, (2) Wrapper methods, and (3) Hybrid methods. The filter methods utilize statistical data-dependent techniques to design optimal channel subsets. These methods are classifier-independent and relatively fast but avoid the relevance of dimensions while selecting channels.

On the other hand, the wrapper methods rely on machine learning algorithms to find an optimal solution from a set of feasible solutions. These methods utilize the classifier's ability to evaluate the quality of selected channels. The wrapper methods are computationally expensive compared to filter methods and results in the overfitting issue. Ultimately, hybrid methods inherit the merits of both the filter and the wrapper methods. Also, these methods are efficient in computing dependencies between channels with lower computational costs than the wrapper methods since they don't compute optimal solutions iteratively. These approaches often exploit different information-theoretic paradigms to measure the importance of channels.

The primary objective of channel selection is to filter only task-specific electrodes that can effectively reduce the computational cost and maximize the classification accuracy of the BCI model. Recently, a variety of algorithms have been proposed to determine the optimal channel subset. For instance, Yang *et al.* [69] computed the most related channels by jointly considering mutual information between Laplacian derivatives of power features extracted from the selected channels and the candidate channels. They also evaluated the redundancy between the power features of the respective channels. The Minimum Redundancy and Maximum Relevance (MRMR) and the Mutual Information-based Channel Selection (MICS) methods efficiently computed the relevancy but were limited by excessive redundancy; theyby ignoring the relevance of channels individually.

Garcia *et al.* [60] developed a fuzzy system interface to obtain a Pareto front solution for classification accuracy maximization. In this problem, a bi-objective function with two criteria (error rate and the number of channels) was used to obtain a robust trade-off between the number of channels and the classification accuracy. In recent work, the Multi-objective Non-Sorting Genetic Algorithm (MO-NSGA) algorithm is used for channel selection [48]. This method employed a hybrid signal feature set using Empirical Mode Decomposition (EMD) and Discrete Wavelet Transform (DWT) with MO-NSGA algorithm and achieved 100% classification accuracy.

Zhang *et al.* [71] improved the performance of the Common Spatial Pattern (CSP) filter bands by decomposing EEG data into spectrum-specific segments. Each of the spectrum-specific segments was further divided into multiple subseries using a sliding window algorithm. They used a joint sparse optimization approach to extract the most discriminating CSP features from each EEG subseries and obtained an average of 88.5% classification accuracy on 60 channel-based BCI datasets.

Chang and Yang [17] applied Stockwell transform and Bayesian linear discriminant analysis in feature generation and classification, respectively, and a Genetic Algorithm (GA) in the channel selection process to find optimal channel subsets. The algorithm was demonstrated on the BCI Competition III dataset I and realize 98% classification accuracy with a 96% sensitivity score. Wang *et al.* [63] reduced the irrelevant and redundant EEG channels using a threshold-based Normalized Mutual Information (NMI) measure. They constructed an NMI connection matrix to obtain the relationship between pairs of channels. Then, setting an appropriate threshold, optimal channels were selected for classification purposes. However, this method achieved better classification accuracy but ignored the relevance of channels individually.

Jiao *et al.* [31] developed an improved CSP variant to capture shared salient information across multiple related spatial patterns using a multiscale optimization approach. To do so, they combined multi-view learning-based sparse optimization to jointly extract robust CSP features with $L_{2,1}$ -norm regularization method and achieved competitive results compared to original CSP and other state-of-the-art methods. Alyasseri *et al.* [5] developed a hybrid channel selection approach using a hybrid flower pollination algorithm and hill-climbing for EEG-based person identification. They utilized three groups of EEG channels (i.e., time domain, frequency domain, and time-frequency domain) in signal extraction. To evaluate the performance of the proposed method, five selection criteria: (1) Channel Reduction Rate (CRR), (2) Classification Accuracy (CA), (3) F-score, (4) sensitivity, and (5) specificity were considered. The proposed approach achieved 96% classification accuracy on a real-world dataset of 109 persons with 14 cognitive tasks using 64 channels.

Zhang *et al.* [71] suggested a clustering-based multitask learning framework to describe the inherent structure of EEG data for achieving more precise MI-related classification accuracy. Further, an affinity propagation decomposition scheme was used to decompose data from each original class into multiple clusters. Finally, a set of CSP features were used to discriminate different clusters corresponding to MI-tasks. The proposed approach achieved a maximum classification accuracy of 87.8% on 60 channel-based BCI datasets. Shi *et al.* [55] introduced a new Binary Harmony Search (BHS) algorithm to find the optimal channel sets and improve the classification accuracy. In this approach, the new harmony was improvised by the existing Harmony Memory Consideration Rule (HMCR) and pitch adjustment operator. The Sparse Representation-based Classification (SRC), Support Vector Machine (SVM), and Linear Discriminant Analysis (LDA) were used on CSP features for MI classification. The test results reveal that BHS achieved better classification accuracy in less computation time than the steady-state genetic algorithms [58].

III. PROPOSED METHODOLOGY

In this section, the entire pipeline of the proposed algorithm for EEG channel selection is discussed. Fig. 2 represent the same through a block diagram. It mainly consists of four steps: (1) EEG data acquisition and trial extraction, (2) signal preprocessing, (3) channel selection & feature engineering, and (3) classification. A detailed description of all the steps is given in subsequent subsections.

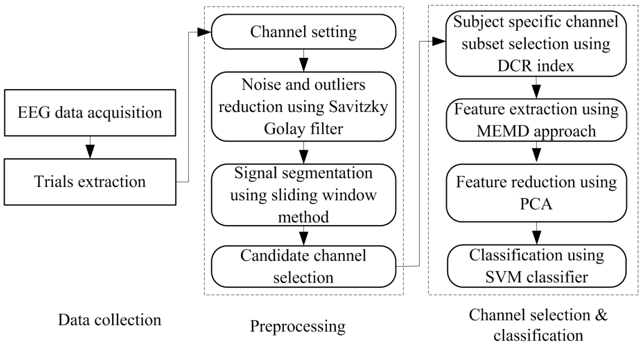

FIGURE 2. Block diagram of proposed EEG channel selection methodology using dynamic channel relevance index.

A. DATA ACQUISITION & TRIAL EXTRACTION

In our study, three EEG datasets from publicly available BCI competitions are employed to validate the effectiveness of the proposed channel selection method: dataset 1 consists of fewer electrodes (22 channels), dataset 2 presents a moderate number of electrodes (59 channels), and dataset 3 is densely packed with the highest number of electrodes (118 channels). A short description of each is discussed in the next subsection.

1) DATASET 1 (BCI COMPETITION IV- 2008 – II A)

This dataset comprises EEG signals collected from 9 healthy participants. This spectrum consists of 22 EEG channels and 3 EOG channels with the left mastoid as the reference. It is a four-class MI task-based dataset where class 1 represents the left-hand movement, the right-hand gesture constitutes class 2, class 3 comprises motion of both feet, and class 4 deals with the tongue activity. This dataset consists of individual training and validation EEG samples for all nine subjects to corroborate any classification scheme. Hence, there is no need to decompose given data samples into the training and the validation sets using any cross-validation technique. The data was recorded from each participant in two different sessions having 6 runs per session (12 runs for the individual participant). Each run consists of 48 trials (12 for each class), which estimates 288 trials in a single session. The dataset was recorded with Ag/Agcl electrodes implanted with an inter-electrode distance of 3 centimeters. All the extracted signals were sampled with a 250Hz frequency using a bandpass filter to collect frequencies between 0.5 Hz to 100 Hz. Throughout the session, the amplifier frequency was fixed to 100μV. Additionally, a notch filter was also applied to attenuate the band spectrum below 20Hz. All processed data is restored in General Data Format (.gdf extension) as one file per subject and session. The experimental setup of each trial is shown in Fig. 3. Dataset 1. More details about this dataset can be found in the reference article [14].

2) DATASET 2 (BCI COMPETITION IV- DATASET 1)

This dataset was recorded from 7 healthy participants through 59 EEG channels. In the entire session, MI-task were performed without any feedback. During the experiment, two participants perform left-hand (L) and foot (F) MI tasks while the rest accomplished the same motion with the right-hand (R) and the left-hand. In the first two runs, the visual cues are shown as left, right, or down keys. The cues were shown on the screen for 4 seconds, during which the subjects are asked to execute respective MI tasks. These periods are shown in the center of the computer screen interleaved with 2 seconds of a blank screen and 2 seconds with a fixed cross. All recordings were collected by using Ag/AgCl electrode cap. The calibration data with a sampling rate of 100 Hz consist of two runs, each of which contains 100 single observations. In our study, the first run of the calibration data was divided into training and testing samples of size 80 and 20 trials, respectively. The experimental setup of each trial is shown in Fig. 3. Dataset 2. More details about this dataset can be found in the reference article [59].

3) DATASET 3 (BCI COMPETITION III - DATASET IVa)

In this dataset, EEG signals are recorded from 5 participants (“aa”, “al”, “av”, “aw”, “ay”) through 118 channels. Each participant executes three MI tasks (left-hand, right-hand, right-foot), but data of only two (left-hand, right-hand) is used in the classification. The visual cues are shown for a period of 3.5 seconds. For all participants, a total of 140 trials are collected and sampled with 100 Hz frequency. Initial 110 trials were used for training, and the rest were utilized in the evaluation session. The experimental setup of each trial is shown in Fig. 3. Dataset 3. More details about this dataset can be found in the reference article [11].

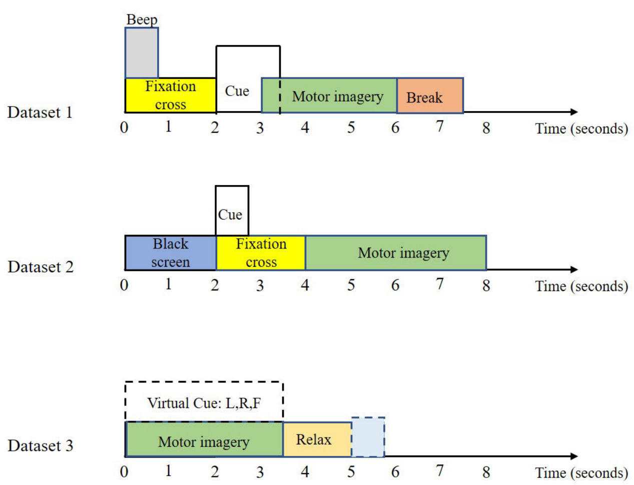

FIGURE 3. The timing sequence of BCI experiments when only the motor imagery section from each dataset is used for feature extraction.

B. SIGNAL PREPROCESSING

In this study, multiple data transformation steps are applied to prepare the raw data for further analysis and classification. These steps strongly concentrate on SNR improvement by reducing the dispersion level of signals and artifacts. These steps are described in detail in this sub-section.

1) EEG CHANNEL SETTING

The primary EEG dataset used in our work consists of a set of 25 EEG channels. Since these channels are Electrooculography (EOG) channels, they are not used for analysis. The data collected from these channels are considered artifacts; therefore, these channels are directly deleted from the dataset, and the remaining 22 channels are used for further processing. In Fig. 4, all 22 EEG channels are topographically shown on the human scalp.

2) SIGNAL FILTERING

Since the SNR of the EEG signals is quite poor and heavily depends on different muscle movement-based artifacts, it must minimize these artifacts before application. Here, we applied the fifth-order Savitzky-Golay bandpass filter [49] of range [0.5-60 Hz] to reduce the induced noise. Bandpass filtering of EEG data reduced a huge amount of irrelevant information and reproduced clean data.

3) SIGNAL SEGMENTATION

After signal filtering, a sliding window method [30] is applied to decompose EEG trials into equal size segments of 0.5 seconds. Since the sampling frequency was 250 Hz, we obtained 125 instances of given MI sequences for further processing.

4) CANDIDATE CHANNEL SELECTION

The optimal channel selection process revolves around the neighborhood of the MI-specific activation region. Therefore, it is required to consider one or multiple EEG channels as reference points to determine the ideal solution. With a reference point-guided method, it becomes simple to find an effective solution using a local search method [61]. MI-based BCI systems generally use three EEG electrodes C_3, C_4, and C_z as candidate channels, which record important properties of MI-specific electrical activity [8], [33], [73], and [51]. The motivation behind the selection of these three electrodes is the neurophysiological basis of the human brain [21], [22]. The motor cortex in the cerebrum is the most sensitive to electrical activities to produce muscle movement. Generally, the motor cortex is located in the frontal lobe and is equally distributed in the left and the right hemispheres of the human brain [43]. MI-based BCI systems use this basis to select candidate channels for recognizing the true level of performed limb activity. During MI execution, C_3 and C_z cover electrical activities generated from the left and right hemisphere, respectively while C_4 collects MI signals from the central part of the human brain. For visualization, the locations of candidate channels in dataset 1 have been shown with dark circles in Fig. 4.

C. CHANNEL SUBSET SELECTION

As earlier discussed, two important factors: (1) channel relevance and (2) channel redundancy, are concentrated over as the objective functions for designing optimal channel subset. In this paper, we developed Dynamic Channel Relevance Index (DCRI) for objective function optimization. The proposed approach is inspired by the Dynamic Feature Importance (DFI), a popular feature reduction method proposed by Wei *et al.* [65] considers Gini Importance (GI) and Maximum Information Coefficient (MIC) for measuring channel relevance and redundancy, respectively. A detailed description of the proposed channel selection method is presented in the following subsections.

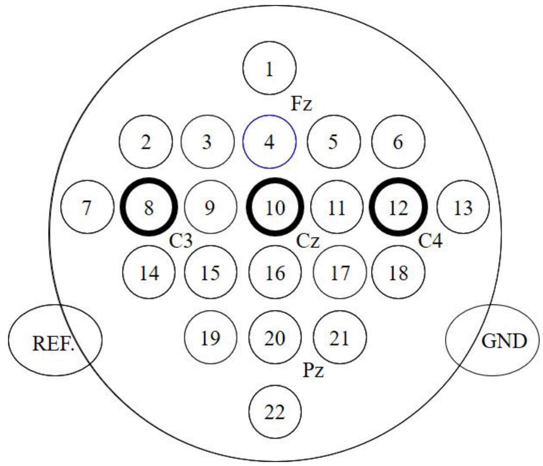

FIGURE 4. Topographical visualization of 22 EEG channels in dataset 1 after channel setting where the dark boundary electrodes are used as candidate channels in our experiment.

1) INFORMATION-THEORETIC BACKGROUND

In information theory, the uncertainty or entropy associated with a n-dimensional random variable $X = \{x_1, x_2, ..x_n\}$ is defined as follows:

$$H\,(X) = -\sum\nolimits_{i=1}^{n} p\,(x_i) \log p(x_i) \tag{1}$$

where $p\,(x_i)$ shows the probability of x_i in X. If the Eq. 1 is extended to two variables X and $Y = \{y_1, y_2, ..y_n\}$, the joint and conditional entropy will be defined by:

$$H(X, Y) = -\sum\nolimits_{i=1}^{n} \sum\nolimits_{j=1}^{m} p\,(x_i, y_i) \log p(x_i, y_i) \tag{2}$$

$$H(X|Y) = -\sum\nolimits_{i=1}^{n} \sum\nolimits_{j=1}^{m} p\,(x_i, y_i) \log p(x_i|y_i) \tag{3}$$

where $p(x_i, y_i)$ shows the joint probability of x_i and y_i, $p(x_i|y_i)$ indicates the conditional probability of x_i given y_i. Mutual Information (MI) represented by $I(X; Y)$ shows the shared information between variables X and Y. It can be defined as:

$$I(X; Y) = \sum\nolimits_{i=1}^{n} \sum\nolimits_{j=1}^{m} p(x_i, y_i)\{\log x_i - \log y_i\} \tag{4}$$

MI and entropy have the following relation:

$$I\,(X; Y) = H\,(X) + H\,(Y) - H\,(X, Y) \tag{5}$$

$$= H\,(X) - H(X|Y) \tag{6}$$

$$= H\,(Y) - H(Y|X) \tag{7}$$

In the case of a third variable Z, conditional mutual information is the mutual information between two other random variables X and Y, which can be given by Eq. 8.

$$I\,(X;Y|Z) = H\,(X|Z) + H\,(X|Y,Z) \tag{8}$$

2) CHANNEL REDUNDANCY MEASUREMENT

Suppose $C = \{c_1, c_2, .., c_n\}$ is the global channel set, $C = \{C_1, C_2, \ldots C_k\}$ is the set of candidate channels, and $S(S \subseteq F)$ is the optimal channel subset. Considering the basic feature selection criteria defined by Wei *et al.* [65], the channel redundancy and relevancy can be defined accordingly (1-4).

- *Definition 1 (Channel Redundancy):* Let c_i and c_j $(i \neq j)$ are two channels in set C, the relationship between both is defined by channel redundancy. Specifically, the degree of redundancy between them is defined by Maximal Information Coefficient-MIC (c_i, c_j). The detailed approach of MIC estimation is given in Algorithm 1.
- *Definition 2 (Strong Redundancy):* Channel c_i is more redundant to channel c_k than channel c_j if

$$MIC\,(c_i, c_k) > MIC(c_j, c_k) \tag{9}$$

where c_k is a candidate channel, and c_i, and c_j are selected channels.

Algorithm 1 Algorithm Pseudocode of MIC Framework $M(c_1, c_2)$

1. Partition scatter plot of c_1 and c_2 over-ordered integer pair of (a, b).
2. To compute the maximum possible mutual information $I^*(c_1, c_2, a, b)$, prepare a grid of size $(a \times b)$ applied to the scatter plot.
3. Normalized the largest mutual information values $I^*(c_1, c_2, a, b)$ such that $M_{a,b}(c_1, c_2) \in [0, 1]$ using Eq. 11.

$$M_{a,b}(c_1, c_2) = I^*(c_1, c_2, a, b)/\log_2 \min(a, b) \tag{10}$$

4. *MIC* is the maximum of $M_{a,b}(c_1, c_2)$ over the ordered pair of (a, b).

$$MIC\,(c_1, c_2) = max\; M_{ab}\,(c_1, c_2)$$

3) CHANNEL RELEVANCE MEASUREMENT

Let C_s be any selected channel from the set C, the role of C_s to candidate channel C_c is defined by channel relevance. Specifically, the relevance of C_s is given by *Gini Index- GI*(C_s).

- ***Gini Index (GI)*** is an important information-theoretic paradigm based on Classification and Regression Tree (CART) [53]. In CART, the purity of dataset D at a given node can be estimated by using *Gini Index* [39] defined in Eq. 11.

$$Gini\,(p_1, p_2, \ldots, p_k) = \sum\nolimits_{j=1}^{k} p_j(1-p_j) \tag{11}$$

where $p_1, p_2, \ldots, p_k$ are the probabilities of class 1, 2. . . , k, respectively. The criterion for CART is to maximize the Gini Decrease (GD). When a node is split according to the channel C_s, the GD of the channel C_s is defined as:

$$GD\,(C_s) = Gini\,(D) - \sum\nolimits_{v} \frac{|D^v|}{D} Gini\left(D^v\right) \tag{12}$$

where D^v is the dataset at the split child node, and CART is generally a binary node. The second term represents the weighted sum of the Gini index at the split child nodes, and its value is always lesser than the parent node.
- *Definition 3 (High Channel Relevancy):* Channel c_i is more irrelevant to channel c_k than channel c_j iff

$$GI\,(c_i) > GI(c_j) \tag{13}$$

where c_k is a candidate channel, and c_i, and c_j are selected channels.
- *Definition 4 (Dynamic Channel Relevance):* The dynamic channel relevance of any channel C_i in the C is estimated as given in Eq. 14.

$$DCR(C_i) = GI\,(C_s) \times \prod\nolimits_{C_x \in C} [1 - MIC(C_i, C_j)] \tag{14}$$

Initially, when the channel subset S is empty, *DCR* of C_s is equal to *GI*. The pseudocode of the DCR algorithm is given in Algorithm.

Algorithm 2 Algorithm Pseudocode of the DCR Algorithm

1) [➢] **Input:** A training Sample D with channel set $C = \{C_1, C_2, \ldots, C_k\}$ and class c and two predefined threshold k and acc.
2) [➢]**Output:** The selected channel subset S

1. $S = \emptyset$
2. $K = 0$
3. $a = 0$
4. ***for*** $i = 1$ *to* k ***do***
5. Calculate the channel importance $GI(C_i)$
6. ***for*** $j = i + 1$ *to* k ***do***
7. Calculate $MIC(C_i, C_j)$
8. **end** ***for***
9. **end** ***for***
10. ***while*** $k < K$ **or** $a < acc$ ***do***
11. Select a channel C_h with the largest $DCR\,(C_h)$
12. $S = S \cup \{C_h\}$
13. $F = F$-$\{C_h\}$
14. Calculate classification accuracy obtained by features extracted from S and record it as a
15. $K = k + 1$
16. **end** ***while***

D. FEATURE EXTRACTION

The channel selected in the previous section is used to extract MI-specific signal properties for the classification of motor imagery activities. Here, we have used the Multivariate Empirical Mode Decomposition (MEMD) method [24] for fragmenting EEG signals. MEMD is an emerging feature extraction approach that has been successfully demonstrated on non-linear and non-stationary EEG signals in numerous BCI systems. It is a multivariate extension of the conventional Empirical Mode Decomposition (EMD) algorithm [24], where multiple n-dimensional envelopes are produced by projecting EEG instances along every direction in n-variate spaces. These projections are further used to compute the local mean. Compared to CSP, MEMD has achieved a superior classification performance in less EEG channel-based BCI models [9], [26]. It indirectly limits the application of the CSP algorithm when the number of selected channels explicitly determines the classification accuracy. In contrast, MEMD can effectively capture the cross-channel dependency and be applied directly to all the EEG channels. Since channel selection is an optimization problem and maintaining a proper balance between selected channels and classification accuracy is our primary objective, we preferred MEMD over other popular feature extraction methods.

Let a multivariate EEG segment be represented by n-dimensional vectors $\{\{x(t)\}_{t=1}^{n} = \{x_1(t), x_2(t), .., x_n(t)\}$ where $d^{\theta_k} = d_1^k, d_2^k, \ldots, d_n^k$ denotes a set of direction vectors along the directions given by angles $\theta^k = \{\theta_1^k, \theta_2^k, \ldots, \theta_{n-1}^k\}$ on an (n − 1) space. The steps used in MEMD computation are given below:

1. Select appropriate points for sampling on (n-1) planes.
2. Compute projection $\{p^{\theta_k(t)}\}_{t=1}^{T}$ along the direction d^{θ_k} of the input signal $\{x(t)\}_{t=1}^{n}$ for all k resulting $\{p^{\theta_k(t)}\}_{k=1}^{K}$ to form a projection set.
3. Find the time series instants $\{t_i^{\theta_k}\}$ corresponding to the maxima of $\{p^{\theta_k(t)}\}_{k=1}^{K}$.
4. Interpolate $\{t_i^{\theta_k}, x(t_i^{\theta_k})\}$ to obtain multivariate envelop curves $\{e^{\theta_k(t)}\}_{k=1}^{K}$.
5. Compute the mean function $m(t)$ by averaging all multivariate envelope curves, defined as follows:
$$m(t) = \frac{1}{n}\sum_{k=1}^{K} e^{\theta_k}(t) \tag{15}$$
6. Compute the detail using $c(t) = Y(t) - m(t)$. If the stopping criterion is satisfied, this detailed Intrinsic Mode Functions (IMF) becomes Multivariate Intrinsic Mode Function (MIMF). Otherwise, $Y(t)$ is assigned to the remainder $c(t)$ and the process to identify a new IMF is reiterated. The entire process is iteratively performed to compute all IMFs from the signal $Y(t)$.

Let $x_i \in R^K$ denote the EEG signal observation at a time instance j where K is the total number of optimal channels. The formal definition of the spatial covariance matrix is defined by $Cov = E\{(x_j - E\{x_j\})(x_j - E\{x_j\})^T\}$, where E. denotes expected value and $(.)^T$ is superscript that represents the transposition of (.). In the BCI system development, each entry of the spatial covariance matrix is considered a feature of the respective observation.

The decomposed EEG segments of N observations are considered to extract segment-wise MEMD features. In Fig. 5, the top nine MIMFs computed from the five channels of subject 1 of BCI competition IV-2008-IIA have been shown. If the size of the covariance matrix is given by (m, n) and the total number of channels is N, then the number of extracted features will be $N \times m \times n$. In the experiment, we obtain a total of 9724 (22*221*2) MEMD features from 22 channels.

E. FEATURE SELECTION

MEMD results in a large feature space because of the numerous decomposition of the EEG signals extracted from the selected channels. To achieve better classification results, it is required to process only those significant features and effectively contribute to discriminate MI tasks. A popular feature reduction method called Principal Component Analysis (PCA) [2] is used to reduce the size of the large feature set using their covariance score. In PCA, the feature matrix F is used to compute the orthogonal matrix $W_{K\times M}$ which is further used to produce the transformed matrix $Y_{K\times N}$ using Eq. 16:

$$Y_{K\times N} = W_{K\times M} \times F_{M\times N} \tag{16}$$

where M is the size of the feature set, N is the number of instances, and $K \leq M$ represents the dimension of the output feature set where K is selected based on the accumulation of the first few largest eigenvalues exceeding 98.6% of the total sum of all the eigenvalues computed from the feature covariance matrix. The first few smallest K values that satisfy Eq. (17) is selected as the model for enumerating the number of the principal component.

$$\frac{\sum_{i=1}^{K}\lambda_i}{\sum_{i=1}^{C}\lambda_i} \times 100 \geq 98.6 \tag{17}$$

Finally, we selected the top P features from the original set that could improve the average (avg.) classification accuracy compared to the absence of feature reduction. In Table 1, we compared the avg. Classification Accuracy (CA) computed on Left vs. Right-hand test data of dataset 1 in two cases: (1) without, (2) with PCA implementation. This comparative analysis indicates that the implantation of PCA on extracted MEMD features plays a vital role in improving the avg. classification accuracy.

F. FEATURE NORMALIZATION

Normalization refers to the scaling of a signal within a predefined range. In this work, we scaled heterogeneous EEG signal instances to significantly compare the classification algorithm's performance over each class of BCI data. Further, a min-max normalization [4] threshold ([0:1]) is fixed such that each instance value falls in the defined data range. Therefore, the min and max values of instances from each

TABLE 1. Comparison between with and without PCA implementation on avg. classification accuracy computed from test samples of dataset 1.

Subject	W/O PCA implementation classification accuracy (in %)	With PCA implementation classification accuracy (in %)
1	59.18	71.33
2	61.42	67.24
3	49.02	63.12
4	68.81	79.82
5	72.20	75.50
6	51.60	68.66
7	69.39	76.80
8	58.26	71.39
9	75.21	80.02
Avg.	62.78	72.65

EEG sample are extracted and then scaled down using the following formula (Eq. 18):

$$x' = \frac{\mathrm{x} - \min(\mathrm{x})}{\max(\mathrm{x}) - \min(\mathrm{x})} \tag{18}$$

where x is a real signal instance at a time, 't' and x' is the scaled value of that particular instance at a time 't'.

G. CLASSIFICATION

In this study, the Support Vector Machine (SVM) [3], [34], [63] is used in MI classification to improve the accuracy, stability and to minimize the overfitting problem. The SVM aims to select a hyperplane in an N-dimensional feature space that can effectively discriminate among the observations of different class labels. Therefore, an SVM classification model maps training samples to a set of data points in space to maximize the gap between two different categories. Let a training sample $S \in R^d$ from two MI classes could be represented as $WS + b = 0$ where $W \in R^d$ is a weight vector and b is the bias. The fundamental goal of the SVM is to find the most optimal hyperplane by solving the following optimization problem

$$\begin{aligned} &minimize\ \frac{1}{2}\,||w||^2 + C\sum\nolimits_n \xi_n \\ &\text{subject to } t_n\left(W^T\Phi(y_n) + b\right) \geq 1 - \xi_n \\ &\qquad\qquad \xi_n \geq 0 \end{aligned} \tag{19}$$

where y_n is training data vector, $t_n \in [1, 2]$ is the class information and n indicate the number of training trials. ξ and C are slack variable and regularization parameters, respectively. In the SVM classification problem, $\Phi(.)$ is employed to map an input data instance into a higher dimensional space using different kernel functions $K(x, y)$. Here, we used Radial Basis Function (RBF) because of its ability to recognize non-linear separability between a given pair of data instances. For SVM classification, the ThunderSVM toolbox [66] was used.

H. PERFORMANCE CRITERIA

The performance of the proposed channel selection approach is evaluated in terms of the following two measures.

1) CHANNEL REDUCTION RATE (CRR)

Channel reduction rate is an indicator that shows how effectively the relevant channels are selected [25]. It is an important measure that reduces the BCI framework's hardware complexity without compromising the classification accuracy. The CRR in this study is computed using Eq. 20.

$$\mathrm{CRR} = 1 - \frac{\text{Number of selected channels}}{\textit{Total number of channels}} \tag{20}$$

2) AVERAGE CLASSIFICATION ACCURACY (ACA)

It is an important measure that describes a classifier's ability to discriminate between different samples using a selected optimal channel set [25]. The classification accuracy can be calculated using Eq. 21.

$$\mathrm{ACA} = \frac{1}{M}\sum\nolimits_{i=1}^{M}\frac{1}{N}\sum\nolimits_{j=1}^{N} match(C_j, L_j) \tag{21}$$

where M is the total number of iterations the algorithm has been executed, N represents the total number of observations in the test dataset, C_j and L_j indicate predicted and true class labels, respectively, and match is a comparison function that provides output 1 when both labels are the same and 0 when different.

IV. EXPERIMENTAL RESULTS

With the aforementioned three BCI competition datasets, extensive experiments are performed to compare the performance of the proposed method with various baseline methods. The performance of our method on each dataset is presented next.

A. TEST RESULTS ON DATASET 1

BCI competition IV- 2008-IIA is a four-class MI dataset; therefore, a multiclass classification approach is required to obtain true class labels from the input feature set. However, the SVM model was meant to perform only binary classification because of the way it creates a hyperplane between two classes. Therefore, we extend it to resolve a multiclass classification problem by applying the One vs. All strategy [41], [45] on the selected MEMD features. No cross-validation scheme is applied during the performance evaluation because training and testing samples are explicitly provided in this dataset.

To investigate the proposed channel selection method's performance, we analyze the relationship between the optimum number of channels and the corresponding classification accuracy. In the learning phase, six MI-task pairs: left vs. right (L vs. R), left vs. foot (L vs. F), left vs. tongue (L vs. T), tongue vs. feet (T vs. F), tongue vs. right (T vs. R), and right vs. feet (R vs. F) were created for each subject, and an SVM classifier is applied over each possible pair of

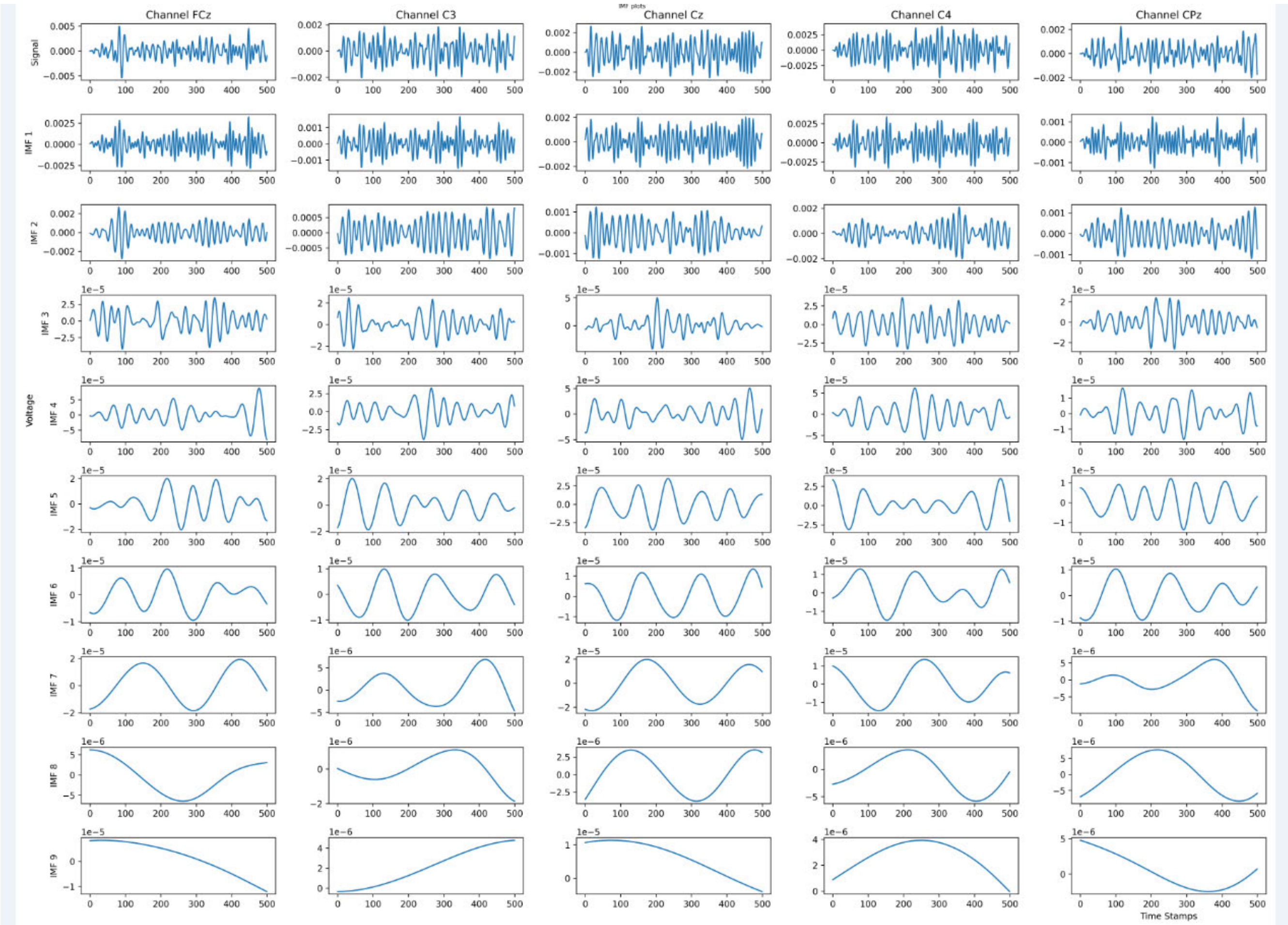

FIGURE 5. The EEG signals and their MIMFs corresponding to the channels FCz, C3, Cz, C4, and CPz for left hand MI task of A01.

MI tasks. Further, the trained classifier is used to classify all unknown test trials of the corresponding train datasets.

We execute our algorithm on all nine subject-specific evaluation samples and calculate mean classification accuracy for all $^4C_2$ MI task pairs. Fig. 6 shows the feature distribution or covariance of the top five left-hand and right-hand MI tasks based on MEMD features. It depicts the boxplot for the MEMD features for the C3 candidate channel estimated by the Kruskal- Wallis test [46]. These five features are statistically significant, with the p-value <0.05 in the training session for the left-hand and the right-hand MI tasks. The feature ranking is performed on the extracted features with the Wilcoxon test [20]. Then, all the features are arranged based on their rank in the descending order of the class separability. A similar approach is also applied for the tongue and foot classes with relative *p*-values of 0.0421, 0.0032, 0.0330, 0.0582, 0.0326. The set of selected normalized features with significant p-values is further used to classify all the MI tasks using the SVM classifier.

Table 2 provides the subject-wise performance of the proposed channel selection scheme in terms of average classification accuracy and channel reduction rate. It can be noticed that maximum classification accuracy is achieved when the proposed channel selection scheme discards approximately 50% of the irrelevant channels. It shows that the proposed channel selection scheme effectively balances the trade-off between the redundancy and relevance level associated with the raw EEG channels. It can be noticed that the average classification performance achieved by the proposed approach is maximum for the R vs. F MI task pair.

In Fig. 7, the effectiveness of the optimal channel subset is computed in terms of the average GI score and size of the optimal channel subset for individual subjects. Initially, the Gini information of the channel subset depends on candidate channels which varies when a new one is added to the subset.

It can be observed that the dynamic relevance of a newly selected channel (C_i) mainly depends on the maximum information that it shares with priory selected channels (C_j).

Moreover, if there is a strong redundancy between the C_j and C_i, the DCR (C_i) will decrease significantly; otherwise, it will increase. Deleting a redundant channel from the global channel set (G_{cs}) will reduce the sharable information among C_j and deleted channel and therefore no change will

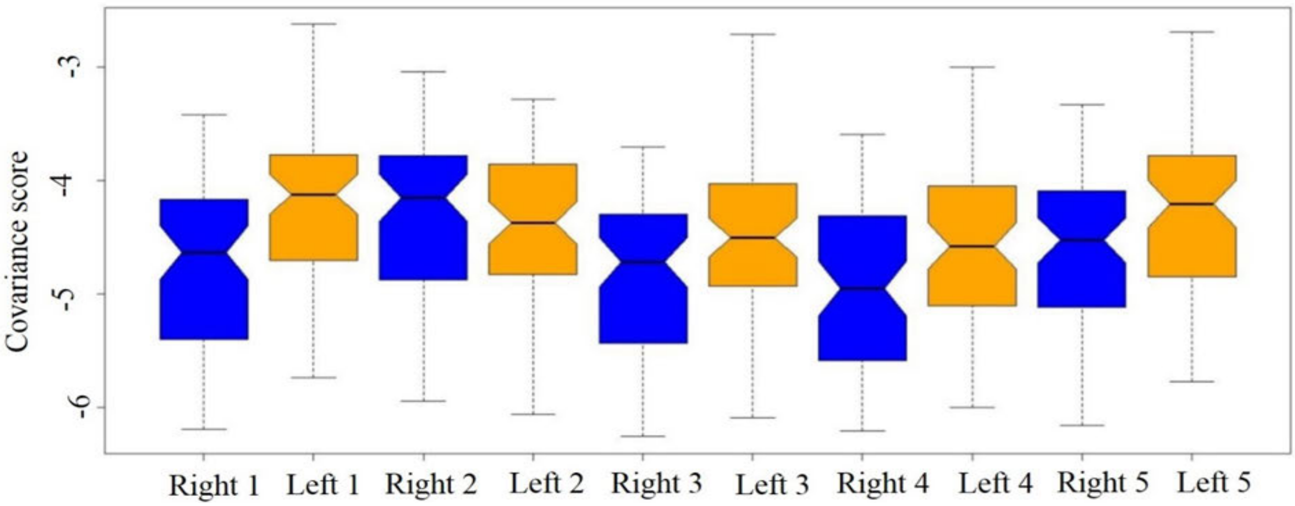

FIGURE 6. Kruskal-Wallis test on top five features from the left hand (orange color) and right hand (blue color) MI task obtained after the applied MEMD approach. The similarity between the second feature of the left and the right hand can be interpreted as a resemblance between the corresponding cognitive control commands. The remaining pairwise (1-1, 3-3, 4-4, 5-5) left-hand and right-hand movement features show a high variance gain.

occur in the GI score of G_{cs}. On the other hand, the selection of redundant channels will increase the sharable information among all the channels and provide a non-zero value of $MIC(C_i, C_j)$, reducing the GI value. Similarly, the selection of non-redundant channels will reduce the sharable information among the channels, maximizing the global channel's GI score. The maximum Gini information of 0.19 is achieved during the experiment for A03, A05, and A07, with a channel subset size of 11 channels. Since EEG signals are subject-specific, computing a global DCR threshold for the total population is impractical.

B. RESULTS COMPARISON WITH BASELINE METHODS

In this subsection, BCI classification results on dataset 1 are compared with two types of published results: (1) publications without and (2) publications with channel optimization. All 22 channels are used in classification in the former, while in the latter, a specific search criterion filters a relevant set. A detailed description of both classes of comparison is given below.

1) PUBLICATIONS WITHOUT CHANNEL OPTIMIZATION

The reported results in Table 2 confirm that the proposed channel selection scheme helps to achieve a maximum classification accuracy of 97.12% (R vs. T) for the eighth participant with an average classification accuracy of 87.72%. Similarly, the other five cases also achieved superior classification accuracy compared to earlier published results where no channel optimization was used. Table 3 shows the comparison of the proposed channel selection scheme with other state-of-the-art methods described in Gaur *et al.* [24]. Method-1 demonstrates the classification results obtained by first executing time window-based bandpass filtering in the range of 8-30 Hz, and then extracted spatio-temporal features were classified using the Riemannian classifier. The proposed method improved classification results by 9.19%, and five out of nine participants have shown improved classification accuracy. Method-2 shows the classification accuracy in the training session using the Subject-Specific Multivariate Empirical Mode Decomposition (SS-MEMD) with CSP features. In this case, we obtained a 9.71 improvement in the classification accuracy across all nine subjects. Method-3 used the CSP features on bandpass-filtered EEG in the range of 8 to 30 Hz. Then, the log-variance of extracted features was computed, and transformed features were classified by Linear Discriminant Analysis (LDA) [37]. Method-4 and 5 used CSP features and identified the covariate shift followed by adaptive and transductive learning, respectively [52]. In method-6, a feature covariance matrix is constructed using the SS-MEMD method, and the Riemannian classifier was used to discriminate four MI tasks. The proposed method improved classification by 7.39 compared to method-5.

2) PUBLICATIONS WITH CHANNEL OPTIMIZATION

In Table 4, the obtained classification results are compared with two published results: (1) Sparse Common Spatial Pattern (SCSP) based channel selections method and (2) Channel set optimization using Improved Binary Gravitation Search Algorithm (IBGSA) [25]. In the SCSP approach [6], two sparse filters were used to maximize weight sparsity among the channels selected by the CSP projection matrix.

Further, two selection criteria: (1) minimum noise and channels irrelevance with utmost classification accuracy, and (2) selected channel minimization while maximizing classification accuracy were used for optimization. Finally, the SCSP algorithm with the first and the second selection criteria are, respectively, abbreviated as SCSP1 and SCSP2. The authors reported that the proposed channel selection

TABLE 2. Performance of the proposed channel selection approach when applied in the classification of six MI tasks pair on test data.

Subject	Classification accuracy of all MI pairs						Selected channels using DCR measure Three candidate channels + (Selected channels)	Number of channels	CRR score
	Lvs. R	Lvs. T	L vs. F	R vs. T	R vs. F	F vs. T			
A01	89.79	87.48	84.79	87.84	91.31	71.08	C_3 ,C_4, C_z, 7, 11, 14, 19, 21	8	0.63
A02	94.18	91.40	94.12	77.88	79.68	87.16	C_3 ,C_4, C_z, 1, 5, 11, 13, 16, 20	9	0.59
A03	78.92	93.30	89.44	91.52	88.44	78.10	C_3 ,C_4, C_z, 2, 4, 7, 11, 15, 16, 19, 22	11	0.50
A04	94.01	79.17	72.68	86.28	87.42	71.44	C_3 ,C_4, C_z, 4, 5, 9, 11, 16, 17, 19	10	0.55
A05	71.32	74.31	83.29	94.42	96.28	93.64	C_3 ,C_4, C_z, 2, 4, 7, 9, 11, 15, 16, 18, 20	12	0.45
A06	86.71	88.51	72.49	87.29	78.33	81.91	C_3 ,C_4, C_z, 1, 4, 6, 9, 14, 16, 18, 19	11	0.50
A07	89.36	73.55	93.18	84.72	93.71	89.24	C_3 ,C_4, C_z, 2, 4, 5, 9, 13, 15, 17, 20	11	0.50
A08	82.11	96.45	85.19	97.12	87.58	82.28	C_3 ,C_4, C_z, 3, 5, 9, 11, 14, 17, 18, 22	11	0.50
A09	86.18	77.30	87.40	81.09	91.76	77.59	C_3 ,C_4, C_z, 2, 6, 9, 11, 15, 16, 17	10	0.55
Avg.	85.84	84.60	84.73	87.72	88.27	81.38	Highly voted channels: 11, 9, 16, 4, 2, 5 15, 17, 19	11	0.53

approach achieved avg. classification accuracy of 81.63% with SCSP1 and 79.07% with SCSP2. In recent work, Improved Binary Gravitation Search Algorithm (IBGSA) automatically detected more informative channels in left or right-hand classification. Here, time and wavelet features were extracted to discriminate four MI tasks using an SVM classifier. The results confirm that IBGSA based channel selection method achieved a maximum of 80% and an average of 76.2% classification accuracy. The results reported in Table 4 show that the proposed channel selection scheme is more effective than both state-of-the-art methods. It improved classification accuracy by 5.1% and 8.5% when compared with SCSP1 and SCSP2 approaches, respectively. Comparing with IBGSA based channel selection approach, the classification results improved by 12.6%, and six out of nine participants have shown enhanced classification results without attenuating avg. classification accuracy.

We also compared the effect of channel reduction rate on subject-wise classification accuracy, as shown in Fig. 7. To do so, a new performance variable, Channel Importance Factor (*CIF*) is calculated using Eq. 22.

$$CIF_i = \frac{Classification\ accuracy\ of\ i^{th}\ participant}{total\ number\ of\ selected\ channels} \quad (22)$$

Here, the CIF_i variable measures the role of each selected channel to obtain classification accuracy. For example, our method achieved 89.79% classification accuracy using eight channels for subject 1; therefore, Channel Importance Factor for subject 1 (CIF_i) will be 11.22 (89.79/8). It can be observed that a channel selection approach will exhibit higher CRR if it has maximum CIF_i value for the i^{th} subject. Fig. 8 showed that the proposed DCR methodology consists of a maximum *CIF* for subjects 1,4,6, and 7 compared to the other three algorithms. Similarly, the IBGSA and SCSP2 achieved the highest *CIF* for subjects 3, 5, and 2, 8, respectively. The SCSP1 obtained the best *CIF* for subject nine compared to other approaches.

We can conclude that the proposed channel selection approach effectively removes redundant and irrelevant information from original EEG channels without attenuating classification accuracy. To evaluate the correctness of classification results, two performance metrics: (1) Sensitivity (SE),

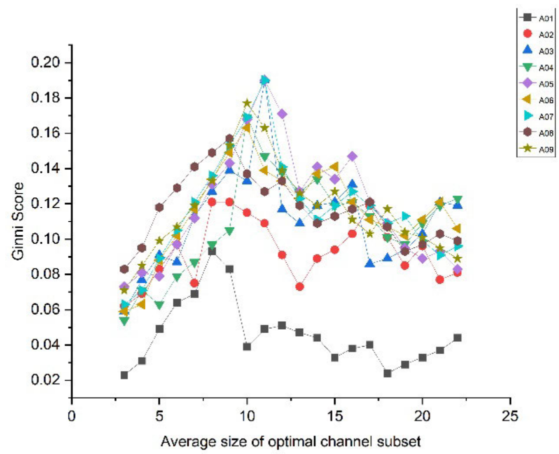

FIGURE 7. A relationship visualization between optimal channel subset and Ginni score.

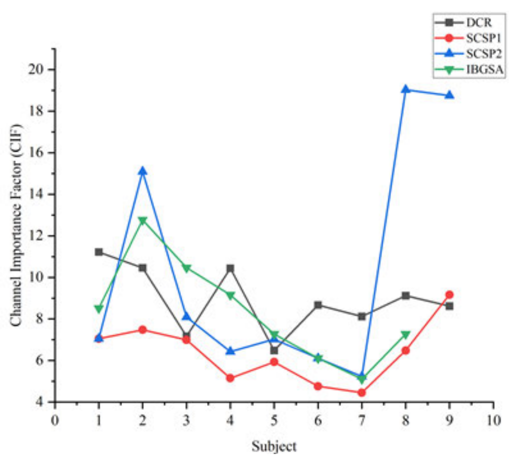

FIGURE 8. Subject-wise comparative analysis of channel importance factor on dataset 1.

TABLE 3. Comparison of classification accuracy (in %) for L Vs. R task of the proposed channel optimization method with other published results (without any optimization) on BCI competition IV dataset 2A. All the results (Method-1, . . . ,6) are taken from Gaur *et al.* [24].

Participants	Proposed method (L vs. R)	Method-1	Method-2	Method-3	Method-4	Method-5	Method-6
				Lotte & Guan, 2011	Raza et al. 2015		Gaur et al., 2018
1	89.79	91.49	92.91	88.89	90.28	90.28	91.49
2	94.18	61.27	59.86	51.39	54.17	57.64	60.56
3	78.92	94.89	95.62	96.53	93.75	95.14	94.16
4	94.01	75.86	66.38	70.14	64.58	65.97	76.72
5	71.32	60.00	62.96	54.86	57.64	61.11	58.52
6	86.71	70.37	68.52	71.53	65.28	65.28	68.52
7	89.36	65.00	68.57	81.25	62.5	61.11	78.57
8	82.11	96.27	96.27	93.75	90.27	91.67	97.01
9	86.18	92.31	93.08	93.75	85.42	86.11	93.85
Avg.	85.84	78.61	78.24	78.01	73.84	74.92	79.93
p-value	0.2478	-	0.7831	0.8147	0.0025	0.0116	0.4241

TABLE 4. Comparison of classification accuracy (in %) for L vs. R task of the proposed channel optimization method with other published results (with optimization) on BCI competition IV dataset 2A.

Participants	Proposed method		Arvaneh et al. 2011 (SCSP1)		Arvaneh et al. 2011 (SCSP2)		Ghaemi et al. 2017 (IBGSA)	
	Selected channels	L vs. R	Selected channels	SVM	Selected channels	SVM	Selected channels	SVM
1	8	89.79	13	91.66	13	91.66	9	76.66
2	9	**94.18**	9	67.36	4	60.41	6	76.66
3	11	78.92	14	97.91	12	97.14	7	73.33
4	9	94.01	14	72.22	11	70.83	8	73.33
5	11	71.32	11	65.27	9	63.19	11	80
6	10	86.71	14	66.67	10	61.11	12	73.33
7	11	89.36	19	84.72	15	78.47	15	76.66
8	9	82.11	15	97.22	5	95.13	11	80
9	10	86.18	10	91.66	5	93.75	-	-
Avg.	10	85.84	13.22	81.63	8.55	79.07	8.6	76.24
p-value		0.4598		-		0.0368		

and (2) Specificity (SP), were computed using Eq. 23-24.

$$Sensitivity = \frac{TP}{TP + FN} \tag{23}$$

$$Specificity = \frac{TN}{TP + FN} \tag{24}$$

where TP, TN, FP, FN are True positive, True negative, False positive, and False-negative instances, respectively. Based on the selected optimal channels, global test results are shown in Fig. 9 as a radar graph [54]. For the given dataset, the average classification accuracy, SE, and SP over all nine subjects was 85.4%, 82.76%, and 81.34%, respectively. In Fig. 10, we have shown the relationship between the position of selected channels and different cognitive regions using subject-wise topographical plots. It explains that recognition of demonstrated MI tasks involves a different combination of channels. Based on selection frequency (>4), a total of nine channels were considered highly voted channels. It is clear from Fig. 10 that the selected channels are mostly distributed in the human brain's central and parietal regions. Since the parietal region is just below the central lobe (location of candidate channels), it can be concluded that there may be a functional similarity between

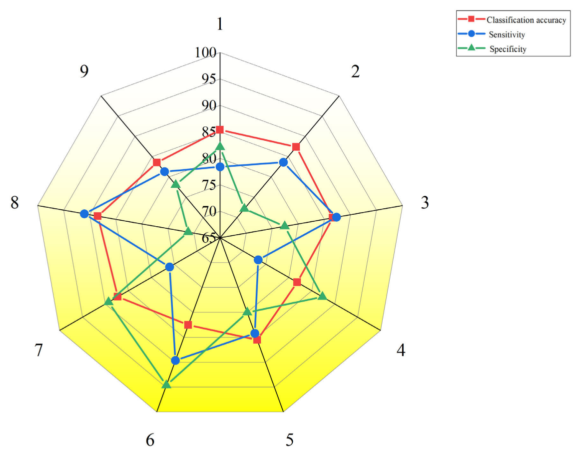

FIGURE 9. Classification accuracy, selectivity, and specificity of all the channels for the BCI dataset.

both cortical regions in terms of motor control and stimuli responses.

C. TEST RESULTS ON DATASET 2 AND 3

We will discuss both jointly in this subsection since the BCI community has provided training and testing data for

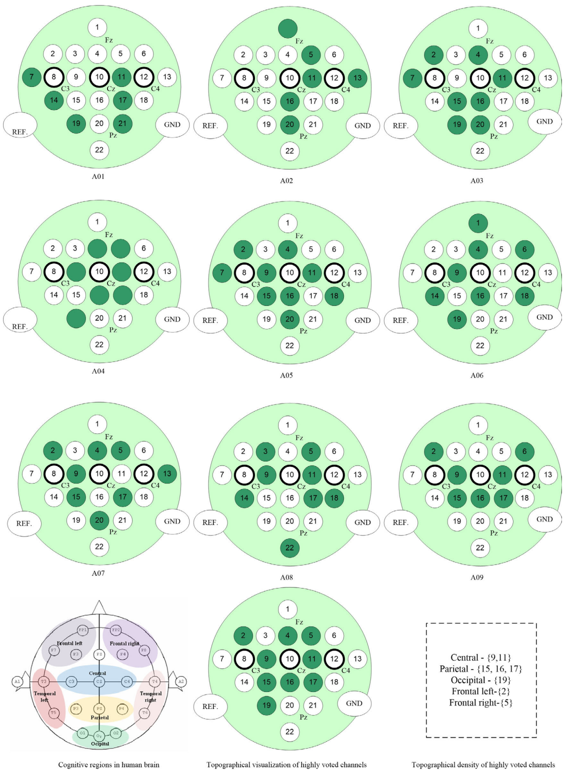

FIGURE 10. Subject-wise topographical visualization of all the selected channels and regional density of highly voted channels in the human brain.

dataset 1 contrary to both datasets 2 and 3. We compared the performance of our method on datasets 2 and 3 with three baseline algorithms: (1) Support Vector Machine Recursive Feature Elimination, (2) Sequential Floating Forward Selection, and (3) Improved Sequential Floating Forward Selection in table 5 and 6. The last row of the tables indicates the *p*-values computed Wilcoxon signed-rank test [20] of the results of ''all channel'' and the channels selected by baseline methods. Support Vector Machine Recursive Feature Elimination (SVM-RFE) [67] is a popular feature selection method employing weight magnitude as a selection norm based on the SVM paradigm. This method selects the feature with the highest-ranking criterion. In the current work, SVM-RFE was employed as a channel selection method, and its results are compared with the proposed method. For each subject, top r ranked channels are selected and used for feature extraction, and finally, the classification accuracy is computed.

Sequential Floating Forward Selection (SFFS) was originally invented by Pudil *et al.* [50]. SFFS selects a feature from the original set that results in the highest objective function value and dynamically eliminates the most useless feature from the previously selected features. Improved Sequential Floating Forward Selection (Improved - SFFS) is a relatively new channel selection algorithm proposed by Qiu *et al.* [51]. They improved the performance of the conventional SFFS algorithm by introducing spatial distribution as an important criterion to select new channels.

In our work, a 5-fold cross-validation scheme [70] is applied on both datasets to avoid overfitting issues. In other words, the datasets are divided into training and testing data samples in the following manner. In the first iteration, 80% of feature vectors are used for training, and the remaining 20% are employed for testing purposes. In the next, another 20% of feature vectors are used for testing, and the rest of the 80% are employed for the training set. This process is repeated until all the feature vectors are used for testing the proposed algorithm. The best classification accuracies from all the participants of datasets 2 and 3 are shown in Tables 5 and 6. For dataset 2, the proposed DCR algorithm achieved an average classification accuracy of 85.59%, significantly higher than the rest of the four methods. Compared to the SVM-RFE, our algorithm yielded an 18% improvement in avg. classification accuracy but used two more channels. However, the proposed method maintained a good balance between the selected channels and achieved avg. classification accuracy compared to the SFFS and Improved SFFS methods. In addition, five out of seven participants obtained the best classification accuracy among all the investigated algorithms. In contrast, the SVM-RFE has shown the best channel reduction rate (CRR) by eliminating more redundant and irrelevant channels.

For dataset 3, the performance of our method is similar to dataset 2. This dataset represents a dense distribution of electrodes. The avg. classification accuracy of the DCR method was 87.57% and yielded an avg. improvement of 5.12%, 4.00%, and 11.41% compared to the Improved SFFS, SFFS, and SVM-RFE methods, respectively, using only 25 channels. The interesting point is that, similar to dataset 2; the SVM-RFE method achieved the best CRR among all the investigated channel selection algorithms.

1) COMPARISON OF CHANNEL REDUCTION RATE

We compared the *CIF* scores of all the channel selection methods to investigate the effect of selected channels on respective avg. classification accuracies. In Fig. 1, the subject-wise CIF scores are compared on datasets 2 and 3. Since the ''all channels'' method did not use any optimization method, it is not included in the comparison. It is clear that the SVM-RFE outperformed other algorithms in the selection of the most informative channels. However, the selected channels were not sufficient to achieve the best classification accuracy. Although, our proposed DCR method selected more channels compared to the SVM-RFE method but achieved the best classification accuracy among all the methods. Therefore, it can be concluded that the DCR method effectively selected informative channels without compromising classification accuracy on both the datasets with a moderate and large number of electrodes. Considering an optimal balance between the selected channels and the classification accuracy, our method can be the best alternative to develop new BCI systems.

TABLE 5. Performance comparison of different channel selection algorithms applied on dataset 2 with 59 channels.

Subject	All channels	Proposed method		Improved SFFS		SFFS		SVM-RFE	
	Accuracy (%)	Accuracy (%)	channels	Accuracy (%)	channels	Accuracy (%)	channels	Accuracy (%)	channels
A	43	75.02	11	69	6	60	9	57	4
B	42	82.80	8	63	15	66	19	54	2
C	62	90.93	21	87	26	91	21	84	32
D	81	97.06	23	94	29	94	34	88	26
E	91	82.8	20	96	19	96	21	95	10
F	49	78.75	7	65	8	58	19	58	4
G	66	91.77	19	72	22	83	21	71	18
Mean	62	**85.59**	15.28	78.0	17.9	78.3	20.6	72.5	13.7
STd	18.9			14.0	8.7	16.5	7.3	16.6	11.8
p-value				0.0025	--	0.002	--	0.0045	--

TABLE 6. Performance comparison of different channel selection algorithms applied on dataset 3 with 118 channels.

Subject	All channels	Proposed method		Improved SFFS		SFFS		SVM-RFE	
	Accuracy (%)	Accuracy (%)	channels	Accuracy (%)	channels	Accuracy (%)	channels	Accuracy (%)	channels
aa	75.7	93.6	24	76.4	27	**78.3**	26	68.6	**7**
al	85.7	79.2	33	**94.3**	47	93.6	43	92.9	**8**
av	62.1	94.6	11	65.0	18	68.6	18	62.9	14
aw	87.1	85.54	26	89.5	27	87.9	**20**	80.0	35
ay	87.1	84.94	31	91.4	35	**92.7**	35	88.6	**18**
Mean	79.5	**87.57**	25	83.3	30.8	84.2	28.4	78.6	16.4
STd	10.9			12.3	10.9	10.6	10.5	12.8	11.3
p-value				0.047		**0.023**		0.75	

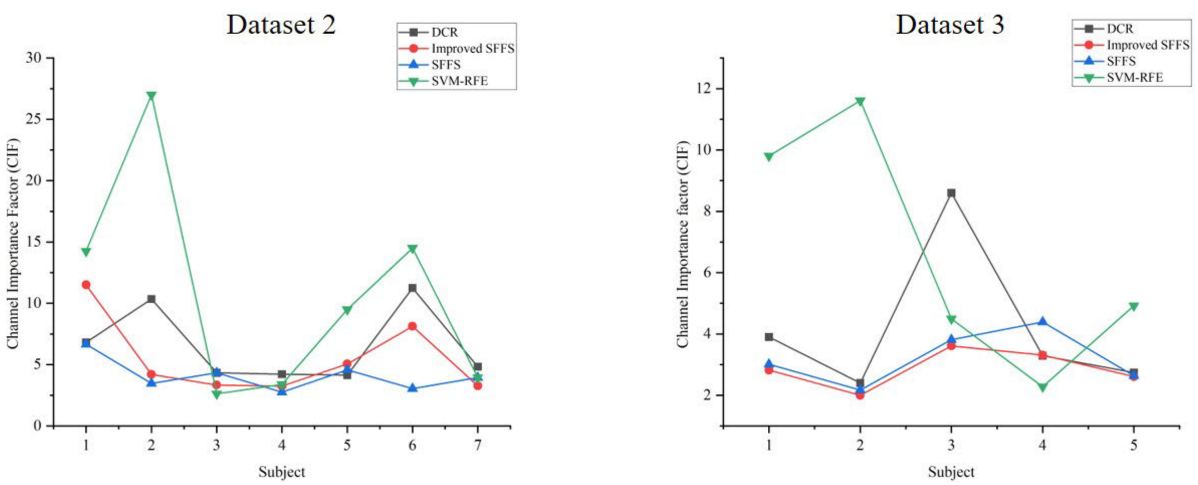

FIGURE 11. Subject-wise comparative analysis of channel importance factor (CIF) scores on datasets 2 and 3.

V. DISCUSSION

A. DISTRIBUTION OF THE SELECTED CHANNELS

The distribution of the channels selected by our method in dataset 2 is shown in Fig. 1. Here, the selected channels are marked with different colors according to their location on the human skull. For three out of seven participants (A, D, and F), the selected channels (shown in the red colors) are equally distributed in the frontal and parietal lobes of the cerebrum while the rest of the participants' (B, C, E, G) selected channels (Yellow color marked) were mainly distributed in the frontal lobe of the human brain. Therefore, it can be concluded that the frontal and parietal cortex significantly contributes to MI activities performance along with the motor cortex (central cognitive region).

B. TIME COMPLEXITY ANALYSIS

In addition to channel reduction rate and classification accuracy, computation time is another important performance indicator of algorithms. The total computation time of the proposed study mainly depends on three steps: (1) channel selection, (2) feature engineering, and (3) classification.

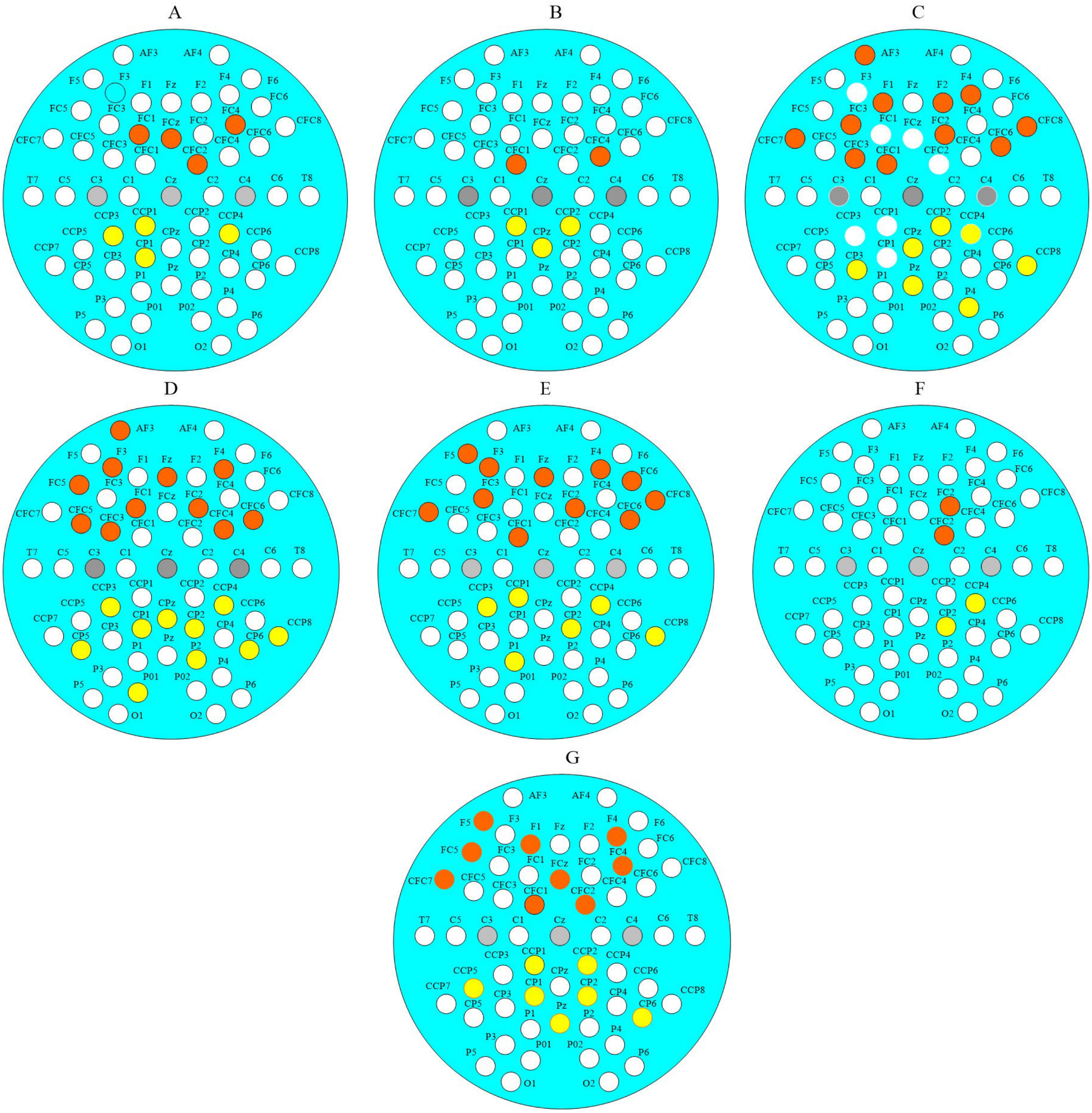

FIGURE 12. Distribution of the selected channels (dataset 2, participants A, B, C, D, E, F, and G). The red circles represent selected channels from the frontal lobe while the yellow-colored circle show the channel distribution in the parietal lobe.

Here, we showed the execution speed of all the algorithms in Fig. 13 on the aforementioned datasets.

All the experiments are performed on a laptop with Intel Core i5 processor-9400F CPU@3.6 GHz; RAM:16G; Windows 10 pr; Matlab 2019b software setup. In Fig.13.A, the operating speed of the proposed DCR method, SCSP1, SCSP 2, and IBGSA on dataset 1 is compared in terms of total execution time in seconds. It can be observed that the computation time of our method is significantly lesser than the rest of the channel reduction methods except subjects 4 and 6. Using the DCR method, the avg. computation time was reduced by approximately 72% (30 seconds) compared to SCSP2. For the rest of the two algorithms, the DCR approach is 57% and 38% faster than SCSP1 and IBGSA methods, respectively.

The time complexity of the four-channel selection methods (All channels, SVM-RFE, SFFS, Improved SFFS) on datasets 2 and 3 is compared with the proposed approach in Fig.13.B and 13.C. It can be observed that the ''all channels'' method is the slowest among the remaining channel

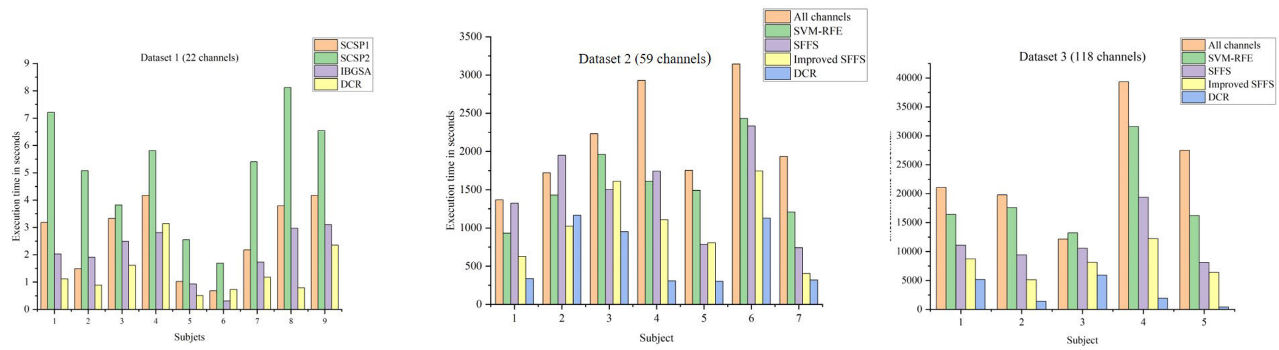

FIGURE 13. Computation time comparison between state-of-the-art channel selection methods on three BCI datasets with varying number of EEG channels.

selection approaches for four participants because redundant and irrelevant data processing is also involved in the MI-based BCI classification. Here, the execution time of all five methods is relatively high compared to dataset 1 because of more number of channels. Using the DCR method, the avg. computation time was reduced by approximately 89% (6890 seconds) compared to the "all channels" method. The execution speed of our method is exceptionally fast compared to the remaining three algorithms for four subjects out of five participants. Compared to the next algorithm, the SVM-RFE approach, our method reduced the computation time by 59.63% (1131 seconds). Similarly, for the rest of the two algorithms, the SFFS and Improved SFFS schemes, our method significantly minimized the avg. computation time by 43.08% (830 seconds) and 27.18%% (429 seconds), respectively.

C. RESULTS IMPLICATIONS

The proposed work aims to improve the classification accuracy of MI-based BCI systems using a minimum number of channels. This alleviates the human effort involved with the BCI preparation setup and reduces the computation burden due to a large number of channels.

In the current research, our results suggest that the frontal and parietal cortex also plays a significant role in the regulation of MI tasks. As we know that BCI systems are used by individuals with motor disabilities and hence facilitate the best possible assistive support. Therefore, the results suggest that our approach may be useful to work on cognitive decline and motor control impairment simultaneously to boost the scope of conventional non-invasive brain-computer interface (BCI) systems. Similarly, our work can be extrapolated to study the Intracranial-EEG (icEEG) [13] behavior of frontal lobe epilepsy [28], [35] where recurring seizures [7] arise in the frontal lobes of the brain. Using the DCR method, an effective classification model can be implemented to distinguish epilepsy from similar disorders such as first seizures [7], febrile seizures [56], nonepileptic events [38], pre-eclampsia [57], meningitis [23], encephalitis [68] since these disorders have similar symptoms and therefore require a detailed clinical history of patients for precise diagnosis. This process is time-consuming and might be erroneous because of manual methods. In such scenarios, our proposed classification model can effectively recognize the true cases with precise classification accuracy.

Similarly, parietal lobe involvement in the MI-task execution also suggests developing advanced BCI hardware modules to facilitate assistance for related neurodegenerative syndromes such as Gerstmann's Syndrome [10] that includes right-left confusion, difficulty with writing (agraphia), and difficulty with mathematics (acalculia). It can also help in recognizing the right parietal lobe dysfunctions such as neglecting self-care skills such as dressing and washing. Some other neuro disorders, such as Balint's Syndrome [27] that also result from bilateral damage of the parietal lobe characterized by the inability to control the gaze (ocular apraxia), voluntarily integrate components of a visual scene (simultanagnosia) and accurately reach for an object with visual guidance (optic ataxia). In medical science, there is no threshold to demarcate the aforementioned parietal lobe disorders from Alzheimer's disease, which is also a progressive neurologic disorder [36]. In this case, a set of the most informative EEG features can be used to construct a hierarchical decision model that can compute the degree of proximity among the diseases, and a set threshold can zero in a particular syndrome.

D. POTENTIAL LIMITATIONS OF THE STUDY

The proposed DCR method yielded higher classification accuracy with a fewer number of channels. The performance of our method is validated on three standard BCI datasets with a different number of electrodes. Despite providing very robust results, our method has the following limitations:

1. The performance of the proposed method is primarily based on the effective link between neurophysiology and the candidate channels. In the case of lacking information from the candidate channels, the performance of our method might degrade.

2. The proposed DCR method selects the relevant set of channels using the Mutual Information Coefficient (MIC) [65]. However, measuring and maximizing mutual

information from finite data is a difficult training objective. Therefore, results may be less informative and error-prone.

3. Our classification model is implemented only on motor imagery datasets. Therefore, its scope is limited only to voluntary limb movements. Extended research is required to find its relevance in similar classification problems such as visual skills, emotion detection, and language interpretation.

4. Although we have validated our approach on three BCI datasets with different EEG channels, the number of participants and dataset size is very limited, resulting in a lack of sufficient persuasiveness. Since neural oscillations are subject-specific and non-deterministic, the proposed method may not provide a global solution.

VI. CONCLUSION

Determining the optimal channel subset is crucial in developing a high-performance BCI system in terms of classification accuracy and computational complexity. In this work, we have developed a novel subject-specific channel selection scheme using Dynamic Channel Relevance (DCR) for the brain-computer interface. Dynamic channel relevance is a recently developed information-theoretic paradigm used for software feature selection. The advantage of the DCR implementation is its ability to investigate effective channels using the ''maximum relevance with minimum redundancy'' principle from a set of raw channels. The proposed method was applied on three different BCI competition datasets with a different number of electrodes to validate the performance. The results indicate the proposed method not only reduces computational complexity but also improves classification accuracy.

The performance of the proposed channel selection scheme is measured using four performance metrics: (1) classification accuracy, (2) channel reduction rate, (3) selectivity, and (4) specificity. Based on the results given in Table 1, Table 2, and Table 3, it can be seen that the proposed channel selection method effectively reduced a large number of channels with a 53% channel reduction rate and overall 85.4% classification accuracy. The sensitivity and specificity of the channel reduction process were 82.76 and 81.34%, respectively, for dataset 1. In addition, the proposed method obtained a set of frequently selected channels ($\geq$4) since they were contributed to improve overall classification accuracy. For example, channel 11 is selected 7 times while channel 9 is selected 6 times. For datasets 2, 3, our method achieved 85.59% and 87.57% avg. classification accuracy, respectively, which is best among all the state-of-the-algorithms. In addition, the CRR of our method is 83% and 78%, respectively. It indicates, proposed channel selection method effectively eliminates irrelevant channels when the size of the dataset increases. In this case, also, the proposed method showed marginal improvement over baseline approaches.

Finally, we have shown the relationship between selected channels and cognitive regions. Since the channel selection process is subject-specific, different sets of channels were found from each participant. Our experiment correlated frequently used channels with different cognitive regions and deduced that MI tasks are mostly distributed in the central, occipital, and parietal regions.

Some prominent applications of proposed methods can be used in different interdisciplinary research fields, such as sequence alignment, where positions of the new acid-base pairs are matched with true ones. Here, the proposed candidate solution-based relevance search approach can be useful to find the importance of new acid-base pairs. Similarly, geostatistical reservoir modeling, weather forecasting, and population ecology are other research domains where the proposed method can be used to find the best solution sets.

Future research may focus on designing hybrid BCI modules, where various cognitive tasks from different brain regions can be effectively decoded simultaneously. It can provide a universal assistive framework for physically disabled patients. In addition, different hybrid methods using metaheuristic algorithms and information theory can be used to search for more effective channels. Some statistical paradigms such as Granger causality and convergent cross mapping can also be used to detect effective channels. Since these methods follow the ''Cause & Effect'' principle, the correlation between relevant channels and classification accuracy can be easily evaluated. Association rule mining is another interesting concept with good scope for finding an effective set of channels and improving classification accuracy. It can be used to design a set of tautological rules to find effective channels.

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